OZARA
SENSOR-ASSISTED OZONE TREATMENT

USER MANUAL

Sensor-assisted ozone treatment unit

Residual odor reduction in dry clothing
For placement inside an enclosed cabinet



SET UP | **OPERATE** | **VERIFY**

Read the safety and ozone-clearance instructions before use. Keep this manual with the prototype.

Prototype edition | October 2026

01 SAFETY AND INTENDED USE

Read before use

WARNING Ozone is a treatment gas

Keep people and animals away from the treatment cabinet and the controlled operating area. Do not inhale cabinet air or use smell to decide whether ozone has cleared.

OZARA is a prototype for residual odor treatment of dry garments. It does not wash fabrics, remove stains, or establish microbial sanitization. Operate only under the documented research or beneficiary operating procedure.

Before every cycle	Required action
Cabinet	Use the assessed enclosure. Check joints, access arrangements, and airflow paths. The portable unit has no cabinet-door interlock.
Garments	Use dry, visibly clean, ozone-compatible items. Check care labels and the assessed loading arrangement.
Electrical supply	Inspect the cable, plug, and housing. Keep hands and the appliance dry. Never open powered electronics.
Clearance equipment	Have the designated ozone-specific meter, its instructions, and the documented access criterion available.

Two requirements before access

The programmed purge must finish, and a separate ozone measurement must meet the documented clearance criterion. A completion sound, green status, or low general gas reading does not authorize cabinet opening.

02 PARTS AND AIRFLOW

Know your appliance



1 Control housing
Controller and power circuits.

2 Uppermost fan
Sensor-sampling airflow.

3 Recessed fan
Ozone circulation, above the OLED.

4 OLED and buttons
Read prompts and use controls.

5 Treatment assembly
Generator and inspection section.

6 Lower fan group
Activated charcoal airflow path.

7 Charcoal drawer
Installed activated charcoal medium.

Keep every opening clear

Do not let garments, cords, bags, or cabinet panels cover the upper sensor fan, recessed ozone fan, side openings, or lower purging fans. Keep the charcoal drawer closed during operation.

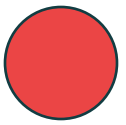
Module dimensions: 29 cm high x 20 cm wide x 22 cm deep.
Mass: 2.75 kg. These describe the portable unit, not the cabinet treatment volume.

03 LOCAL CONTROLS

Buttons and display



BLUE OK



RED MINUS



YELLOW PLUS

Control	Action in the relevant screen
OK: tap	Start empty-cabinet sampling at Ready; continue after loading; choose the suggested duration; acknowledge completion and open Results.
OK: hold 3 seconds	Confirm the selected treatment duration. In Results, return to Ready without unplugging.
MINUS: tap	Choose manual duration or reduce it. In Results, switch result pages. Other screens may switch the reading view.
PLUS: tap	Increase the manual duration.
PLUS: hold 6 seconds	Request cancellation where permitted. During countdown or treatment, this leads to mandatory purging. Purging cannot be skipped.

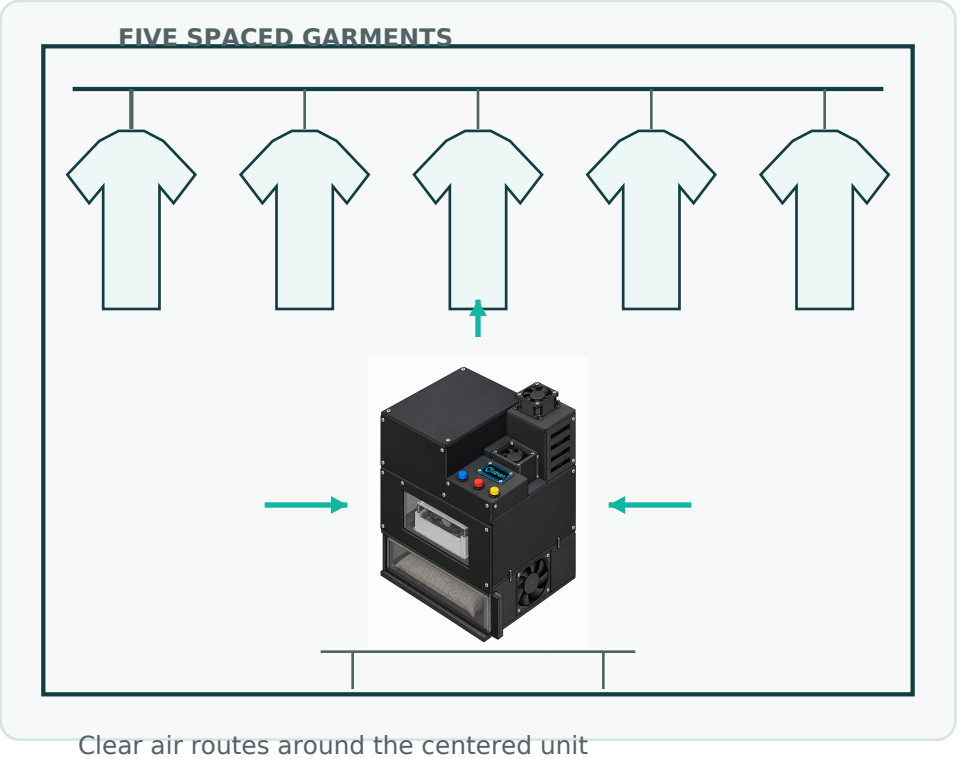
Follow the prompt for the current stage

A button can have different functions in different screens. Keep confirmation holds continuous and release when the action is accepted. Do not press several controls at once.

04 CABINET SETUP

Position and load correctly

Place the prototype at the center of the cabinet on a stable, level support. Arrange up to five dry garments neatly with clear spaces between them and every fan opening.



Clear air routes around the centered unit

Keep hanging hems above the unit and away from its inlets. Do not pack shirts against each other or the cabinet walls. Route the cable so it cannot snag, touch moving fans, or interfere with cabinet closure.

Loading diagram is illustrative

Five garments are an intended arrangement, not a validated treatment capacity. Diagram not to scale; use the assessed cabinet and keep all fan routes clear.

05 STARTUP

Plug in and sample the baseline

1. **Inspect while unplugged**

   Check the housing, cable, fan openings, and closed charcoal drawer. Prepare the designated ozone meter and clearance procedure.

2. **Install in the empty cabinet**

   Place OZARA centrally. Keep the treatment cabinet free of garments for the baseline measurement. Ensure operating controls can be used through the assessed access arrangement.

3. **Plug in to power on**

   Connect the intact supplied power lead to the appropriate mains outlet. The prototype starts when power is connected; there is no separate power-button step.

4. **Wait for the Ready prompt**

   Allow startup to finish and check that readings are usable under the prescribed sensor-readiness procedure. A startup countdown is not proof of complete sensor conditioning.

5. **Tap the blue OK button**

   Follow the empty-cabinet prompt. The sampling fan runs and the controller collects baseline readings. Do not add clothing during acquisition.

6. **Wait for the load-clothes prompt**

   The baseline is stored and the sampling fan stops at the relevant transition. Proceed only when prompted.

If the baseline is invalid

Do not continue through a sensor fault. Check the displayed message, power, sensing arrangement, and acquisition conditions using the troubleshooting guide.

06 PREPARING THE CYCLE

Load and confirm a duration

1. **Load at the indicated stage**

   Use the assessed garment count and arrangement. Keep all fan openings unobstructed, close the cabinet, then tap OK as prompted.

2. **Allow clothing sampling to finish**

   Odor buildup, fan preparation, and clothing sampling follow the programmed sequence. Do not reposition garments during these stages.

3. **Review the recommendation**

   The displayed suggestion is calculated from sensor changes. It is provisional and does not indicate measured ozone concentration or proven odor removal.

4. **Choose suggested or manual duration**

   Tap OK to select the suggested option. Select manual adjustment using MINUS, then use MINUS or PLUS to change the time. The inspected controls allow 1-5 minutes with nominal 30-second steps and a stop at the suggested value.

5. **Hold OK for 3 seconds**

   Review the selected value, then hold OK continuously to confirm. The countdown begins before ozone generation. Keep the cabinet closed and the operating area controlled.

Duration is not continuous generator ON time

The selected interval contains pulsed generation and generator-off circulation. Do not interpret the displayed minutes as an ozone concentration, exposure dose, or guaranteed treatment result.

07 AUTOMATIC OPERATION

Treatment and mandatory purging

Stage	What to expect
Countdown	Outputs follow the programmed delay. Keep the cabinet closed and do not reach inside.
Pulsed treatment	The circulation fan operates while the generator switches ON and OFF according to the programmed pulse schedule.
Generator-off circulation	The generator stops while circulation continues for the remaining selected interval.
Purging	The generator remains OFF and the lower fan group moves air through the activated charcoal path. Cancellation cannot bypass this stage.
Complete / Results	The programmed sequence has ended. A sound and display prompt may request acknowledgement; clearance is still a separate requirement.

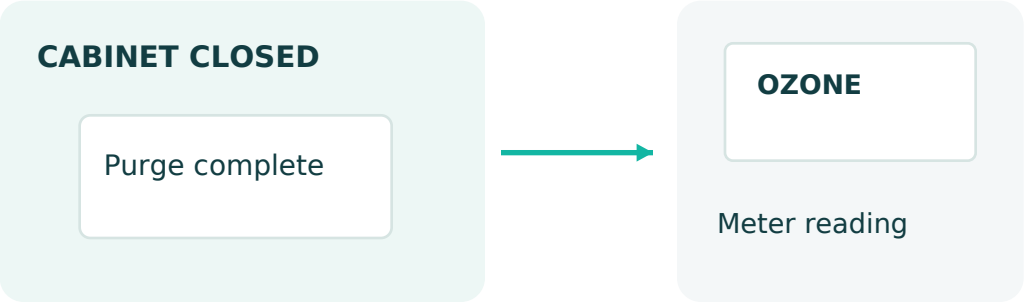
Do not unplug to stop an ordinary cycle

Unplugging also stops powered circulation and purging. Use the supported cancellation action where appropriate and let the required purge finish. Handle power loss as an interrupted cycle.

Cancellation: hold PLUS for 6 seconds where supported. Before treatment, cancellation returns to the appropriate starting stage. During countdown, pulsed treatment, or generator-off circulation, the generator is switched off and the controller enters mandatory purging. During purging, a cancellation request is rejected.

08 OZONE METER CHECK

Verify clearance before access



1. **Keep the cabinet closed**

   Finish mandatory purging. Do not use the OLED gas readings, a dashboard status, or the absence of odor as an access decision.

2. **Use the designated ozone-specific instrument**

   Check the meter is suitable for the required range and has passed its specified calibration or functional check. Follow its own warm-up and sampling instructions.

3. **Measure through the prescribed sampling arrangement**

   Check the documented internal and external locations without opening the cabinet merely to take a reading. Follow the approved observation period and acceptance criterion.

4. **Release access only when the criterion is met**

   The responsible operator records the result and authorizes opening. If the reading fails or is unavailable, keep access restricted and follow the documented recovery procedure.

Use the designated ozone instrument

Establish the meter, sampling arrangement, numerical clearance criterion, and responsible operator before ozone-enabled operation. A general gas module is not an ozone-clearance instrument.

09 ENDING A CYCLE

Results, shutdown and interruptions

1. **Acknowledge completion**

   At the completion prompt, tap OK to acknowledge the sound and open Results. Use MINUS to switch available result pages.

2. **Keep completion separate from clearance**

   Acknowledge the interface without opening the cabinet. Retrieve garments only after the independent ozone-clearance procedure authorizes access.

3. **Save records before shutdown**

   Download any needed local CSV records and check cloud synchronization. Local records are held in volatile memory and can be lost when power is removed.

4. **Finish or prepare a new cycle**

   For normal shutdown, unplug only after the required purge, clearance, and record-saving steps. Grip the plug rather than pulling the cable. In Results, holding OK for 3 seconds returns to Ready without rebooting.

Power failure or accidental disconnection

Treat the run as interrupted. Keep the cabinet closed and access restricted. The inspected firmware starts with outputs off after restart but does not automatically restore a verified recovery purge. Do not start another ozone-generating cycle as a substitute. A designated operator must complete the documented ozone-disabled recovery procedure and verify clearance.

If smoke, sparking, or an electrical fault is suspected, keep people away and isolate electrical power from a safe position. Arrange qualified inspection before further use; clearance restrictions remain applicable.

10 LOCAL AND FIREBASE ACCESS

Dashboard and recorded data

Interface	Purpose and access
Local ESP32 page	Connect to the unit network identified by the operator. Open the supplied local address; the inspected configuration uses http://192.168.4.1. Use the operator-issued access details.
Wi-Fi setup	Where the installed interface supports it, use the local /wifi page to configure a 2.4 GHz internet connection. Follow the displayed connection status.
Firebase website	Open https://ozara-dfec7.web.app/login.html and sign in with an authorized account. This is separate from the local connection.
Monitoring and history	Check device identity, operating state, timestamps, and whether readings are current. Missing cloud updates do not establish that the physical cycle stopped.
Data export	Save the local CSV before unplugging. Identify which records came from local export and which successfully synchronized to Firebase.

Use verified controls only

Local virtual buttons mirror supported physical controls in the inspected interface. A visible cloud control or status card does not by itself demonstrate remote actuation. The ozone meter remains the basis of clearance verification.

Reading labels: BME680: kΩ; MQ136/MQ137: raw ADC; temperature: °C; humidity: %RH. These are not odor-intensity or ozone-clearance readings.

11 CHECK THE SYMPTOM

Troubleshooting

Symptom	Operator action
No display after plugging in	Check the outlet and visible cable condition from a safe position. Stop using a damaged lead or housing. Do not open powered electronics.
Sensor fault / no valid samples	Stop progression. With the generator off and applicable clearance completed, have the operator inspect the sensor connection and acquisition conditions. Retry only after the cause is addressed.
Empty and loaded readings look similar	Check placement, unobstructed sampling airflow, and the defined measurement procedure. Record the limitation. Do not claim that a similar reading proves the garment is odor-free.
Button does not perform the expected action	Check the displayed state, release other controls, and use the correct tap or continuous hold. Wait for announcements to finish. Purge cancellation is intentionally blocked.
Fan seems stopped or blocked	Do not touch blades or reach into an operating cabinet. Treat airflow failure as a fault and apply the interruption and clearance procedure.
Dashboard is stale or disconnected	Check the correct network, account, device, and timestamps. Observe the local status without bypassing cabinet-access restrictions.
Odor remains after a completed cycle	Record the outcome and loading condition. Do not infer generator failure from odor alone. Any additional treatment must use the documented settings and repeat the full purge and clearance requirements.

12 CARE OF THE PROTOTYPE

Cleaning and maintenance

Before any care task

Complete the applicable clearance procedure, disconnect power, and allow components to cool. Do not open the charcoal drawer or housing during treatment, circulation, or purging.

Item	Care and inspection
Housing and acrylic	Wipe accessible surfaces with a soft cloth. Keep liquids out of electronics, ports, and seams. Avoid abrasives and incompatible solvents.
Fan openings	With power disconnected, remove accessible dust and obstructions. Do not force blades or change fan orientation.
Activated charcoal assembly	Inspect the medium, bag, drawer seating, and airflow path. Follow its replacement procedure; do not assume washability or regeneration.
Sensors	Keep openings clear. Do not spray cleaner or touch sensing elements. Qualified personnel handle replacement or reconditioning.
Power and fasteners	Check cables, supports, panels, and overheating marks. Have electrical defects assessed before reuse.
Maintenance record	Record the date, inspection, fault, charcoal or component change, corrective action, and person responsible.

Store the cleared, unplugged unit upright and dry. Lift by its stable housing and protect the display and fans. Never lift by a fan, drawer, display, or cable.

13 KEEP WITH THE APPLIANCE

Timing and configuration reference

The shared code history identifies the following original timing profile. Later firmware revisions use different stage values. Check the installed configuration and displayed prompts before operation; these entries are not validated treatment or clearance settings.

Original reference stage	Reference setting
Startup settling / fan preparation	5 s / 5 s
Empty and clothing sampling	60 s each
Clothing odor buildup	120 s
Selected treatment interval	60-300 s; nominal adjustment 30 s
Generation / circulation split	First half pulsed; remaining half generator OFF
Generator pulse schedule	ON 5-15 s; OFF 15-30 s; score-dependent
Original final purge	60 s; not an ozone-clearance criterion

Use the installed profile

A different stage timer on the actual unit takes precedence over this original reference. Record its configuration on the next page. Never shorten a purge or treat a reference value as proof of clearance.

14 APPLIANCE RECORD

Operator configuration record

Complete this record before ozone-enabled operation and update it whenever the configuration, meter, cabinet, or approved procedure changes. Keep the supporting records with the prototype.

Installed firmware / date

Approved operating settings record

Ozone meter model / check record

Clearance criterion and procedure ID

Assessed cabinet / loading record

Responsible operator / contact

Sources: supplied prototype videos; approved logo and model images in the manuscript; shared firmware history and inspected project source. This manual does not certify treatment effectiveness or residual-ozone safety.